

Lab: Sculpting Curvature

Objective

Experience curvature as a hands-on topological operation. You will create three surfaces with three different curvatures — flat, hyperbolic (ruffled), and spherical (cupped) — using only increases and decreases as your tools. Then you will shape a small 3D form by combining all three curvature types, and map each shaping decision to the topological operation it performs.

Materials

- Worsted weight yarn in one main color, approximately 40 yards
- Worsted weight yarn in a contrasting color, approximately 20 yards
- Crochet hook, size H/8 (5.0mm)
- Pen and Curvature Log (template below)
- Blank paper for surface diagrams
- Ruler or tape measure
- Stitch markers (4-5)
- Polyester fiberfill stuffing (a small handful)
- Scissors

Curvature Log Template

Round	Stitch Count	Increases/ Decreases	Observed Shape	Curvature Type	Notes

Steps

Part 1: The Flat Disc — Zero Gaussian Curvature (15 minutes)

The goal here is a perfectly flat circle. Flatness requires a precise balance of increases — exactly enough new surface area to fill the expanding circumference, no more, no less.

1. With your main color, make a **magic ring**.
2. **Round 1:** Work 6 sc into the magic ring. Pull the ring closed. Place a stitch marker in the first stitch. (6 sc)
3. **Round 2:** 2 sc in each st around. (12 sc)
4. **Round 3:** *Sc in next st, 2 sc in next st*, repeat around. (18 sc)
5. **Round 4:** *Sc in next 2 sts, 2 sc in next st*, repeat around. (24 sc)
6. **Round 5:** *Sc in next 3 sts, 2 sc in next st*, repeat around. (30 sc)
7. **Round 6:** *Sc in next 4 sts, 2 sc in next st*, repeat around. (36 sc)
8. Fasten off and lay the circle on a flat surface. Does it lie flat? Press it gently. If it ruffles at the edge, your tension was a bit loose (effectively adding extra area). If it cups slightly, your tension was tight.
9. **Record** in your Curvature Log: stitch count per round, increases per round (always 6), observed shape (flat, slight ruffle, slight cup), curvature type (zero).

Part 2: The Hyperbolic Ruffle — Negative Gaussian Curvature (15 minutes)

Now we deliberately break the flat rule. We will add too many increases, forcing the fabric to ruffle into a hyperbolic surface.

1. With your contrasting color, make a **magic ring**.
2. **Round 1:** Work 6 sc into the magic ring. (6 sc)
3. **Round 2:** 2 sc in each st around. (12 sc) — same as the flat disc so far.
4. **Round 3:** 2 sc in each st around. (24 sc) — *doubling* every stitch instead of the flat rule.
5. **Round 4:** 2 sc in each st around. (48 sc) — doubling again.

6. You should already see dramatic ruffling. The fabric cannot lie flat; it waves and folds like a lettuce leaf or a coral frond. This is negative Gaussian curvature — more surface area than a flat plane can accommodate.
7. Fasten off. Try to flatten it against the table. It refuses. It has too much surface for its boundary.
8. **Record:** increases per round (6, 12, 24 — far exceeding the flat rule of 6), observed shape (extreme ruffling), curvature type (negative / hyperbolic).

Part 3: The Bowl — Positive Gaussian Curvature (15 minutes)

Now the opposite: too few increases, so the fabric cups into a bowl.

1. With your main color, make a **magic ring**.
2. **Round 1:** Work 6 sc into the magic ring. (6 sc)
3. **Round 2:** *Sc in next st, 2 sc in next st*, repeat around. (9 sc) — only 3 increases instead of 6.
4. **Round 3:** *Sc in next 2 sts, 2 sc in next st*, repeat around. (12 sc) — again only 3 increases.
5. **Round 4:** *Sc in next 3 sts, 2 sc in next st*, repeat around. (15 sc) — 3 increases.
6. **Round 5:** *Sc in next 4 sts, 2 sc in next st*, repeat around. (18 sc) — 3 increases.
7. **Round 6:** *Sc in next 5 sts, 2 sc in next st*, repeat around. (21 sc) — 3 increases.
8. The fabric should be cupping noticeably — forming a bowl or dome shape. It curves upward from the center because there is not enough surface area for flatness.
9. Fasten off. Place it on the table. It sits like a little cup or lens.
10. **Record:** increases per round (3 — half the flat rule), observed shape (cupping, bowl), curvature type (positive / spherical).

Part 4: Sculpting a 3D Form — The Amigurumi Ball (20 minutes)

Now combine all three curvature types to create a three-dimensional closed surface: a simple ball.

1. With your main color, make a **magic ring**.
2. **Round 1:** 6 sc in ring. (6 sc)
3. **Round 2:** 2 sc in each st. (12 sc)

4. **Round 3:** *Sc, 2 sc in next st*, repeat. (18 sc)
5. **Round 4:** *Sc in next 2, 2 sc in next st*, repeat. (24 sc)
 - **Pause and observe:** You are in the positive-curvature expansion phase. The fabric is cupping into a dome. Mark this moment in your Curvature Log.
6. **Rounds 5-8:** *Sc in each st* around with no increases. (24 sc each round)
 - **Pause and observe:** The fabric is now a cylinder — zero curvature. The sides go straight up. Mark this.
7. **Round 9:** *Sc in next 2, sc2tog*, repeat. (18 sc)
8. **Round 10:** *Sc, sc2tog*, repeat. (12 sc)
9. Stuff the ball lightly with fiberfill.
10. **Round 11:** *Sc2tog* around. (6 sc)
11. Fasten off, leaving a long tail. Thread through the remaining 6 stitches and pull tight to close.
 - **Observe:** The decrease rounds created positive curvature again, closing the surface. You have made a closed manifold — a sphere-like surface with no boundary.
12. **Record** the full curvature journey: expansion (negative boundary, positive curvature) to cylinder (zero curvature) to contraction (positive curvature) to closed surface.

Part 5: Surface Diagrams and Mapping (15 minutes)

1. On your blank paper, draw three simple side-view profile diagrams:
 - **Flat disc:** A horizontal line (zero curvature throughout)
 - **Hyperbolic ruffle:** A wavy line showing the fabric's undulation (negative curvature)
 - **Bowl:** An upward-curving arc (positive curvature)
2. Draw a side-view profile of your amigurumi ball, labeling the three zones:
 - Bottom dome: positive curvature (increase phase)
 - Middle cylinder: zero curvature (straight rounds)
 - Top dome: positive curvature (decrease phase)
3. Next to each zone, write the topological operation that created it:
 - "6 increases/round = expansion"
 - "0 increases/round = cylinder"
 - "6 decreases/round = contraction"

- "Close final hole = seal the manifold"
4. On a separate section of your paper, draw a simple neural network loss curve (U-shape or valley) and label how the crocheter's shaping corrections parallel gradient descent steps toward the minimum.

Discussion Questions

- Compare your flat disc, hyperbolic ruffle, and bowl side by side. All three started from the same magic ring. What single variable (increases per round) determined whether the surface was flat, ruffled, or cupped?
- When you made the amigurumi ball, at what point did the topology change from an open surface (with a boundary) to a closed surface (no boundary)? What was the final operation that closed it?
- If you were to frog the ball back to Round 4 (the end of the increase section), what would the topology of the remaining fabric be? How does this relate to the concept of rolling back gradient steps in neural network training?
- Look at your surface diagrams. How does the crocheter's generative model (the mental image of a ball) guide the transition from increases to straight rounds to decreases? What would happen if the crocheter had no generative model and simply crocheted randomly?

Wrap-Up

You have sculpted three curvatures and one closed 3D form using only increases, decreases, and straight stitches as your tools. Keep your flat disc, ruffle, bowl, and ball — they are physical demonstrations of Gaussian curvature and topological surgery. In the next module, we will explore how the skills you used today were learned in the first place.